RONSHEE CHAWLA

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EDUCATION

- **University of Texas at Austin** (UT)
  PhD in Electrical and Computer Engineering — August 2017 – Present
  MS in Electrical and Computer Engineering — August 2017 – May 2019
  Advisor: Prof. *Sanjay Shakkottai*
  Current GPA: 4.0/4.0

- **California Institute of Technology** (Caltech) — September 2016 – August 2017
  Graduate Student in Electrical Engineering
  GPA: 3.7/4.0

- **Indian Institute of Technology (IIT) Indore**, India — July 2011 – May 2015
  B. Tech. in Electrical Engineering
  Advisor: Prof. *Prabhat K. Upadhyay*
  GPA: 9.89/10.0

RESEARCH INTERESTS

- Design and analysis of algorithms for decentralized/distributed machine learning
- Reinforcement learning theory and algorithms
- Bandit algorithms

PUBLICATIONS (* denotes equal contribution)

- *Collaborative Multi–Agent Heterogeneous Multi–Armed Bandits*
  R. Chawla, Daniel Vial, Sanjay Shakkottai, R. Srikant
  International Conference on Machine Learning (ICML) 2023 (28% accepted, to appear)

- *Multi–Agent Low–Dimensional Linear Bandits*
  R. Chawla, Abishek Sankararaman, Sanjay Shakkottai
  IEEE Transactions on Automatic Control, 2023

- *The Gossiping Insert–Eliminate Algorithm for Multi–Agent Bandits*
  R. Chawla*, Abishek Sankararaman*, Ayalvadi Ganesh and Sanjay Shakkottai
  International Conference on Artificial Intelligence and Statistics (AISTATS) 2020 (30% accepted)

- *Outage Performance and Location Optimization for Traffic–Aware Two–Way Relaying with Direct Link*
  Suneel Yadav, R. Chawla and Prabhat K. Upadhyay
  International Conference on Signal Processing and Communications (SPCOM) 2016

- *Outage Analysis of Cellular Two-Way Relaying with Spectrum Sharing in Nakagami–m Fading*
  R. Chawla, M. Mounika Reddy and Prabhat K. Upadhyay
  Proceedings of 18th International Symposium on Wireless Personal Multimedia Communications (WPMC) 2015

TECHNICAL SKILLS

- **Programming languages**: C++, Python
- **Machine learning frameworks**: PyTorch
- **Softwares**: MATLAB

Work Experience

- **PhD Research Software Engineer Intern** Summer 2022
  Uber Technologies Inc., Sunnyvale, CA USA
  Worked in Marketplace Investments Modeling team, where I proposed a new deep neural network model for optimal allocation of budget across driver incentives to maximize the revenue, in a computationally efficient manner

- **Scientist-B** August 2015 – May 2016
  Aeronautical Development Establishment (ADE), DRDO, Bengaluru, India
  Flight Test Telecommand and Tracking Division

- **Internship** May 2014 – July 2014
  RWTH Aachen University, Germany
  Worked on automatic extraction of layer-specific data-flow configurations for automatic scheduling and mapping of kernels for the Layers architecture

Teaching Experience

- *Probability, Statistics and Random Processes*, Teaching Assistant, Fall 2017
  Online Learning (**Graduate**), Teaching Assistant Fall 2019
  - Duties included conducting office hours, and setting homework and exam problems

Talks

- *The Gossiping Insert–Eliminate Algorithm for Multi–Agent Bandits*
  Invited talk at Vrbo, Austin, TX February 2020

Academic Service

- Reviewer for IEEE Conference on Decision and Control 2022, 2023
- Reviewer for IEEE Control Systems Letters 2022
- Reviewer for Conference on Neural Information Processing Systems (NeurIPS) 2022
- Reviewer for International Conference on Artificial Intelligence and Statistics (AISTATS) 2023

Honors and Achievements

- First rank in Electrical Engineering PhD Qualifying Exam at Caltech, January 2017.
- **President of India Gold Medal**, first rank in IIT Indore, class of 2015.
- DAAD WISE scholarship for summer internship at RWTH Aachen University, Germany, May 2014 – July 2014.
- Offered summer fellowship by Indian Academy of Sciences, May 2014 – July 2014 (declined).
- Academic Excellence Awards, IIT Indore, 2012 – 13, 2013 – 14 and 2014 – 15.

Graduate Coursework

- **Machine Learning**: Learning Systems, Special Topics in Unsupervised Learning (GANs), Statistical Learning Theory, Fair Transparent Machine Learning

- **Mathematics**: Probability and Stochastic Processes, Advanced Probability, Convex Optimization (Theory and Algorithms), Information Theory, Applied Linear Algebra, Random Matrices, Coding Theory, Mathematics of Signal Processing, Queueing Theory